

GLOBAL AIR POLLUTION ANALYSIS AND AQI CATEGORY PREDICTION

Data Analytics, Machine Learning & Power BI Project Report

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1. Executive Summary

This project presents an end-to-end analysis of global air pollution data to understand air-quality patterns and develop a machine learning system for predicting Air Quality Index (AQI) categories.

The workflow covers data understanding, data cleaning, exploratory data analysis, feature engineering, classification modeling, model evaluation, model interpretation, prediction, and Power BI visualization.

The dataset contains 23,463 records and 12 variables describing AQI and pollutant-related measurements for cities around the world. Logistic Regression achieved 90.54% accuracy, Random Forest achieved 98.55% accuracy with 89.94% Macro F1, and Balanced Random Forest achieved 98.32% accuracy with 92.34% Macro F1.

Balanced Random Forest was selected as the preferred final model because it provided better class-balanced performance, particularly for minority AQI categories. Feature-importance analysis identified PM2.5 AQI Value as the strongest reported predictor.

2. Project Objectives

- Understand the structure and quality of the global air pollution dataset.
- Clean missing and incomplete records before analysis.
- Explore AQI and pollutant distributions using statistical summaries and visualizations.
- Prepare features for multi-class AQI category classification.
- Compare Logistic Regression, Random Forest, and Balanced Random Forest.
- Evaluate models using accuracy and Macro F1, with attention to class imbalance.
- Interpret the final model using feature importance.
- Develop a reusable AQI category prediction system.
- Create a Power BI dashboard for interactive analysis.

3. Business Problem

Air pollution is a major environmental challenge, and air-quality conditions vary across locations. A data-driven approach can help identify pollution patterns, understand which pollutant measurements are most associated with AQI categories, and support consistent classification of air-quality conditions.

The project investigates which pollutants influence AQI, how AQI categories are distributed, how accurately machine learning can classify AQI categories, whether class balancing improves minority-class performance, and which variables contribute most to predictions.

4. Dataset Description

The project uses the Global Air Pollution Dataset created by Hasib Al Muzdadid and available through Kaggle. It contains city-level geographic information together with overall AQI and pollutant-specific AQI measurements.

Dataset size: 23,463 rows and 12 columns. Variables include Country, City, AQI Value, AQI Category, CO AQI Value and Category, Ozone AQI Value and Category, NO2 AQI Value and Category, and PM2.5 AQI Value and Category.

5. Data Cleaning

Initial inspection found 427 missing values in Country and 1 missing value in City. The remaining AQI and pollutant measurement columns contained no missing values. The dataset contained 0 duplicate rows. The missing/incomplete records were handled during the cleaning stage, and the cleaned dataset was saved as a processed CSV for later notebooks.

6. Exploratory Data Analysis

EDA examined numerical distributions, potential outliers, AQI category frequencies, pollutant relationships, and correlations.

Histograms were used for numerical variables, box plots for spread and potential outliers, bar charts for AQI categories, and correlation analysis for numerical relationships.

The analysis highlighted the importance of pollutant-related AQI measurements and the imbalance between AQI categories.

7. Feature Engineering

Feature engineering prepared the cleaned data for machine learning by selecting relevant variables, encoding categorical information, and preparing the target for multi-class classification.

Data leakage was treated as an important modeling consideration. Features that directly reveal or are derived from the target can artificially inflate model performance, so the modeling workflow kept the target separate from predictive inputs.

8. Machine Learning Methodology

The prediction task was formulated as a multi-class classification problem with six AQI categories: Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and Hazardous.

Logistic Regression was used as a baseline. Random Forest was evaluated as a nonlinear ensemble method. Balanced Random Forest was evaluated to improve learning when class frequencies are uneven.

9. Model Evaluation

Accuracy and Macro F1 were used to compare models. Accuracy measures overall correct predictions, while Macro F1 gives equal weight to each class and is therefore useful when categories are imbalanced.

Logistic Regression: 90.54% accuracy.

Random Forest: 98.55% accuracy and 89.94% Macro F1.

Balanced Random Forest: 98.32% accuracy and 92.34% Macro F1.

Although Random Forest had slightly higher accuracy, Balanced Random Forest achieved the higher Macro F1 and improved recall for minority classes.

10. Final Model Selection

Balanced Random Forest was selected as the preferred final model. Its 98.32% accuracy remained very high while its 92.34% Macro F1 indicated stronger performance across AQI classes.

For this problem, balanced performance across all classes is important because rare AQI categories should not be overlooked simply because they contain fewer observations.

11. Model Interpretation

Feature-importance analysis showed that PM2.5 AQI Value was the most influential feature, with importance 0.307262. Ozone AQI Value had importance 0.079345, CO AQI Value 0.059108, Country–India 0.039138, and NO2 AQI Value 0.038032.

PM2.5 was therefore the dominant feature in the reported model-importance results, while ozone, CO, NO2, and selected country indicators also contributed.

12. Prediction System

A prediction system was developed in the final notebook using the trained model. It accepts relevant input features and returns a predicted AQI category, with prediction probabilities available for additional confidence information. One evaluated example produced approximately 0.76 probability for Unhealthy and 0.20 for Very Unhealthy, resulting in an Unhealthy prediction.

13. Power BI Dashboard

A Power BI dashboard was created to provide an interactive business-intelligence layer for the project. It enables exploration of overall AQI patterns, AQI categories, pollutant levels, and country/city-level air-quality information. The dashboard complements the Python notebooks by presenting analytical findings in an interactive format suitable for communicating results to non-technical users.

14. Key Findings

1. PM2.5 AQI Value was the most important reported feature.
2. Random Forest achieved the highest overall accuracy at 98.55%.
3. Balanced Random Forest achieved the highest reported Macro F1 at 92.34%.
4. Balancing improved recall for minority categories, including the Hazardous class.
5. Macro F1 provided a more informative view of class-balanced performance than accuracy alone.
6. The project combines descriptive analytics, predictive modeling, model interpretation, and business intelligence.

15. Limitations and Future Improvements

The dataset may not represent every location or time period uniformly.

Model performance depends on the quality and distribution of the available AQI and pollutant measurements.

Class imbalance remains an important consideration for rare AQI categories.

Future work could incorporate time-series information, weather variables, traffic information, and additional environmental features.

Future modeling could compare gradient boosting, calibrated classifiers, and additional explainability techniques.

16. Conclusion

This project demonstrates a complete data analytics and machine learning workflow for global air-quality analysis, from raw data understanding through cleaning, EDA, feature engineering, modeling, evaluation, interpretation, prediction, and dashboard development.

The results show that tree-based methods were highly effective. Random Forest achieved the highest accuracy, while Balanced Random Forest provided the best balance across classes according to Macro F1. PM2.5 emerged as the dominant feature in the model-importance results.

Overall, the project demonstrates practical skills in Python, data analysis, visualization, machine learning, model evaluation, model interpretation, prediction, and Power BI.

17. Project Structure

```
Global Air Pollution Dataset/
├── data/
│   ├── raw/
│   └── processed data/
├── models/
│   └── aqi_category_model.pkl
├── Notebook/
│   ├── 01_data_understanding.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_eda.ipynb
│   ├── 04_feature_engineering.ipynb
│   ├── 05_ml_model.ipynb
│   ├── 06_model_evaluation.ipynb
│   ├── 07_model_interpretation.ipynb
│   └── 08_prediction_system.ipynb
├── PowerBI/
├── images/
├── README.md
└── requirements.txt
```

18. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib, Jupyter Notebook, and Microsoft Power BI.

19. Dataset Reference

Global Air Pollution Dataset, created by Hasib Al Muzdadid and available on Kaggle.

20. Model Performance Summary

Model	Accuracy	Macro F1
Logistic Regression	90.54%	—
Random Forest	98.55%	89.94%
Balanced Random Forest	98.32%	92.34%

21. Notebook Workflow

- 01 — Data Understanding: Load and inspect the dataset.
- 02 — Data Cleaning: Handle missing/incomplete records and prepare clean data.
- 03 — EDA: Explore distributions, categories, outliers, and correlations.
- 04 — Feature Engineering: Prepare predictive variables and target.
- 05 — ML Model: Train and compare classification models.
- 06 — Model Evaluation: Evaluate classification performance.
- 07 — Model Interpretation: Analyze feature importance and model behavior.
- 08 — Prediction System: Generate AQI category predictions.